# Vision Can't Feel: Minimal Pairs for Contact-Dependent Robot Success Judgment, and What an Agent Should Check Instead

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## Abstract

Vision-language models (VLMs) are increasingly used to judge whether a robot finished its task. Recent work found these judges near chance when success depends on contact. We study that failure under control. We generate *minimal pairs* in MuJoCo: two snapshots of the same randomized scene, one a success and one a failure that differs by a few millimeters (touching vs. hovering, grasped vs. beside, seated vs. on the rim, pressed vs. barely pressed, touching a wall vs. short of it). We compare free VLM judges with tiny classifiers on three deliberately degraded, low-cost sensor signals: a fingertip contact switch, gripper effort, and a single-axis wrist force. We also search for the largest physical failure that still fools each VLM. On 400 samples the best free VLM we could run at scale reached 55% balanced accuracy (95% CI [50, 60]) with a 54% false-success rate; a logistic regression on the degraded sensors reached 89% ([85, 92]) with 11%. Sensors fail where they cannot observe the contact (push, 50%) and do not transfer across tasks. Some VLMs still call a failure a success at gaps of several millimeters. Free-tier quotas limited coverage for several models; we report it throughout.

## 1 Motivation

A success detector gates retries, rewards, and data collection. A false success is the costly error: the robot moves on while the task silently failed. For contact-dependent tasks the evidence is often a millimeter gap or a force, which a camera at 640 pixels sees poorly or not at all. Cheap sensors measure contact directly. We ask how far that intuition holds when the sensors are realistically noisy, and where it breaks.

## 2 Related work

FailBench [1] evaluates 13 VLM detectors on 2,197 real and simulated attempts. Detection is near saturation when success shows as object motion and near chance on contact-rich assembly (no model above 0.60 balanced accuracy on its REASSEMBLE subset), with a bias toward predicting success. It uses vision only and names force-torque as future work. Guardian [2], AHA [3] and RoboFAC [4] train vision-language failure detectors from perturbed demonstrations; all are vision-only. Force and touch appear in policies (e.g. force-aware VLAs) rather than as success judges, and classical multimodal anomaly detectors predate VLM judges. To our knowledge no prior work compares cheap non-visual sensors with VLM judges on the same attempts. The vision-side finding is therefore not new; our contribution is the controlled minimal-pair setting, the degraded-sensor comparison, fusion, and the physical attacker. FailBench data was not yet released when we checked, so we could not report on its subset. On the agent side, LLM planners over a fixed skill library with success feedback follow SayCan [5] and Inner Monologue [6]; Inner Monologue already feeds success detection back into the plan. Using touch to decide whether a grasp worked goes back at least to Calandra et al. [8], and acting to perceive (our press probe) is interactive perception [7].

## 3 Setup

**Simulation.** MuJoCo 3.17 with the Menagerie Franka Panda (gravity compensation, position actuators, damped least-squares IK, guarded moves that stop at 8 N of tared wrist force). Five tasks, each with a matched failure: **stack** (held cube touching the base vs. hovering), **grasp** (closed on the cube vs. beside it), **button** (pressed past a 4 mm click vs. less), **peg** (seated vs. on the rim), **push** (block against a wall vs. short of it).

**Ground truth.** Explicit rules on simulator state after a 2 s stability hold with controls frozen: contact force between the right bodies, cap displacement, peg depth plus floor contact, and for grasp a lift test. Images and sensors are captured *before* the hold, at the moment a judge would decide. A pair counts only if its labels come out opposite.

**Randomization.** Seeded per pair: key and fill light, table tint, object colors from a palette, three camera poses (one raised oblique, two low profile views), and 0 to 4 distractors kept out of camera sight lines.

| Task | Pairs | Failure magnitude (mm) | Front pixel diff |
|---|---|---|---|
| stack | 40 | 0.5–2.0 | 3.6 (median) |
| grasp | 40 | 1.1–3.9 | 24.8 (median) |
| button | 40 | 0.4–2.4 | 5.3 (median) |
| peg | 40 | 0.5–3.9 | 10.3 (median) |
| push | 40 | 0.6–3.0 | 3.7 (median) |

Table 1: Scaled set: minimal pairs per task.

**Pairs.** Table 1 shows failure magnitudes and the median front-view pixel difference within pairs.

**Cheap sensors.** From the simulator at 100 Hz: fingertip touch (thresholded at 0.3 N into a binary switch per finger), gripper actuator force (a motor-current proxy), and wrist force along the tool axis only. Degradation per unit strength: switch flips 2%, 30 ms latency; effort noise 0.6 N, 10-bit over ±40 N, drift up to 0.8 N; force noise 0.5 N, 8-bit over ±30 N, offset up to 1.5 N, drift up to 1 N; 2% dropped samples; 20 ms latency. Twelve summary features feed logistic regression and a small gradient-boosted tree, trained on 600 samples from seeds disjoint from every test set.

**VLM judges.** Free tiers only: Gemini 3.5 Flash, Gemini 3.1 Flash-Lite and Gemma 4 31B through Google AI Studio; Qwen 3.8 27B through Groq; Qwen2.5-VL 3B and Moondream 1.8B through local Ollama on a CPU laptop (images downscaled to 384 px for these two). Each gets a task description and one front image (single view) or all three views (multi-view), and returns JSON with a verdict and a confidence. Every answer is cached by (model, image hashes, prompt). Calibration uses the stated confidence.

**Fusion.** Logistic regression on sensor features plus the VLM's P(success), 5-fold cross-validated on the test set with folds grouped by pair.

# 4 Results

**Main comparison.** Table 2 is the scaled set (400 samples). Table 4 adds all models and multi-view on the 100-sample set. Free-tier daily quotas stopped some models early; the Coverage column says how many samples each judge answered, and metrics use only those. On the scaled set the sensor classifier beats every VLM we ran on balanced accuracy and false-success rate (Table 2). The best VLM is gemini/gemini-3.1-flash-lite (single) at 0.55 [0.50, 0.60]; on the 164 samples it answered, a paired McNemar test against the sensor classifier gives $p < 0.001$ (65 samples only the sensors got right, 6 only the VLM got right; Table 3). Fusing it with sensor features gives 0.88 (164/400 samples). On the small set the best VLM is gemini/gemini-3.1-flash-lite (multi) at 0.65 with 57/100 answered; low-coverage rows are indicative only. Vision does best on grasp, where an empty gripper is visible, and worst where the failure is a sub-millimeter gap or a few millimeters of travel.

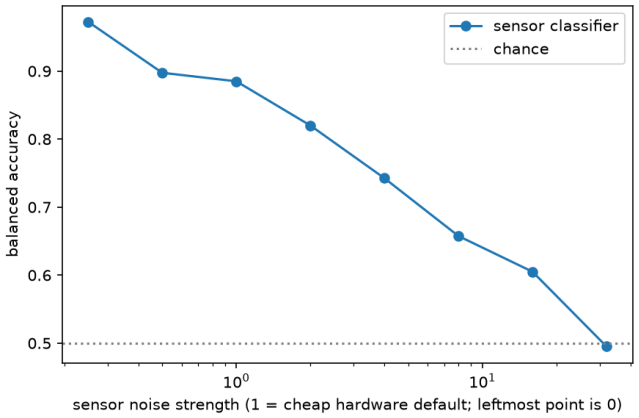



Figure 1: Sensor classifier balanced accuracy vs. noise strength; dashed: best VLM on the same set.

**Where sensors fail.** The single-axis wrist force runs along the tool, so a sideways push barely registers, and the fingertip switch only sees the push if the contact lands inside its zone. Table 5 shows the push column. Sensor classifiers also do not transfer across tasks (Table 6): a model trained on four tasks is near chance on the fifth, because the same signal means different things in different tasks.

**Noise robustness.** Table 7 and Figure 1 scale every degradation together. Accuracy falls from 0.97 with ideal sensors to 0.89 at the default and 0.50 at strength 32. Its 95% interval first overlaps the best VLM's at strength 16, and its point estimate falls below the VLM's at strength 32.

**Calibration.** Figure 2 plots reliability. VLMs report high confidence whether right or wrong, so their ECE is large (gemini/gemini-3.1-flash-lite (single): 0.45); the sensor classifier's ECE is 0.04.

**Millimeters to fool.** For each model, task and scene the attacker climbs a ladder of failure magnitudes (0.5 to 16 mm; 0.25 to 3.7 mm below the click for the button), bisects between the largest fooled and smallest not-fooled value, then runs four LLM-proposed mutations (magnitude, light, camera height) that see all previous results. Every candidate is re-simulated and discarded if ground truth turns into success. Table 8 reports the largest failure still judged a success, averaged over two scenes; Table 9 re-judges every fooling scene with the other model and the sensor classifier. gemini/gemini-3.1-flash-lite: fooled in 2 of 4 scenes, up to 13.5 mm.

| Judge | Coverage | Bal. acc. [95% CI] | False-succ. [95% CI] | False-fail. | ECE |
|---|---|---|---|---|---|
| sensor_gboost | all | 0.89 [0.85, 0.92] | 13% [8, 17] | 10% | 0.06 |
| sensor_logreg | all | 0.89 [0.85, 0.92] | 11% [7, 16] | 12% | 0.04 |
| fusion: gemini/gemini-3.1-flash-lite (single) + sensors | 164/400 | 0.88 [0.82, 0.93] | 10% [4, 17] | 15% | 0.05 |
| gemini/gemini-3.1-flash-lite (single) | 164/400 | 0.55 [0.50, 0.60] | 54% [43, 65] | 37% | 0.45 |
| ollama/moondream:1.8b (single) | 8/400 | 0.50 [0.50, 0.50] | 100% [100, 100] | 0% | 0.17 |

Table 2: Scaled set (400 samples). Sorted by balanced accuracy.

| VLM | Shared | Only VLM right | Only sensors right | $p$ | Test |
|---|---|---|---|---|---|
| gemini/gemini-3.1-flash-lite (single) | 164 | 6 | 65 | <0.001 | chi2_cc |
| ollama/moondream:1.8b (single) | 8 | 0 | 3 | 0.250 | exact |

Table 3: Paired McNemar test of each VLM against the sensor classifier on the samples the VLM answered (scaled set).

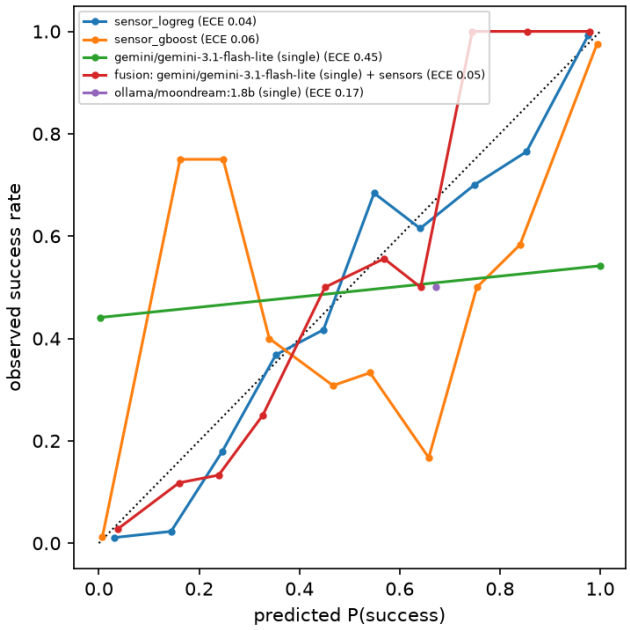



Figure 2: Reliability diagrams on the scaled set. VLM confidence is self-reported.

ollama/moondream:1.8b: fooled in 10 of 10 scenes, up to 16.0 mm. The sensor classifier called 6 of 115 fooling scenes a success.

| Judge | Coverage | Bal. acc. [95% CI] | False-succ. [95% CI] | False-fail. | ECE |
|---|---|---|---|---|---|
| fusion: gemini/gemma-4-31b-it (single) + sensors | 83/100 | 0.87 [0.79, 0.94] | 8% [0, 17] | 19% | 0.10 |
| sensor_logreg | all | 0.86 [0.78, 0.93] | 14% [6, 24] | 14% | 0.06 |
| sensor_gboost | all | 0.86 [0.78, 0.93] | 12% [4, 22] | 16% | 0.10 |
| fusion: ollama/moondream:1.8b (single) + sensors | 100/100 | 0.86 [0.79, 0.93] | 8% [2, 16] | 20% | 0.10 |
| fusion: gemini/gemini-3.1-flash-lite (multi) + sensors | 57/100 | 0.86 [0.76, 0.95] | 10% [0, 21] | 18% | 0.12 |
| fusion: gemini/gemini-3.1-flash-lite (single) + sensors | 100/100 | 0.85 [0.77, 0.92] | 8% [2, 16] | 22% | 0.13 |
| fusion: ollama/qwen2.5vl:3b (single) + sensors | 100/100 | 0.85 [0.77, 0.92] | 10% [2, 18] | 20% | 0.10 |
| fusion: groq/qwen/qwen3.8-27b (single) + sensors | 93/100 | 0.84 [0.75, 0.91] | 11% [2, 19] | 22% | 0.11 |
| fusion: gemini/gemini-3.5-flash (single) + sensors | 41/100 | 0.78 [0.65, 0.90] | 24% [10, 43] | 20% | 0.18 |
| gemini/gemini-3.1-flash-lite (multi) | 57/100 | 0.65 [0.55, 0.74] | 45% [28, 62] | 25% | 0.35 |
| gemini/gemini-3.5-flash (single) | 41/100 | 0.61 [0.48, 0.74] | 52% [29, 76] | 25% | 0.34 |
| gemini/gemini-3.1-flash-lite (single) | 100/100 | 0.60 [0.54, 0.66] | 46% [32, 60] | 34% | 0.40 |
| groq/qwen/qwen3.8-27b (single) | 93/100 | 0.57 [0.49, 0.65] | 43% [28, 57] | 43% | 0.36 |
| gemini/gemma-4-31b-it (single) | 83/100 | 0.56 [0.48, 0.64] | 38% [23, 53] | 51% | 0.42 |
| gemini/gemma-4-31b-it (multi) | 9/100 | 0.55 [0.30, 0.80] | 40% [0, 80] | 50% | 0.43 |
| ollama/moondream:1.8b (single) | 100/100 | 0.50 [0.50, 0.50] | 100% [100, 100] | 0% | 0.23 |
| ollama/qwen2.5vl:3b (single) | 100/100 | 0.50 [0.50, 0.50] | 100% [100, 100] | 0% | 0.44 |

Table 4: Small set (100 samples), all judges including multi-view. Partial coverage reflects free-tier quotas.

| Judge | stack | grasp | button | peg | push |
|---|---|---|---|---|---|
| sensor_gboost | 1.00 | 1.00 | 0.98 | 1.00 | 0.46 |
| sensor_logreg | 0.94 | 1.00 | 1.00 | 0.99 | 0.50 |
| fusion: gemini/gemini-3.1-flash-lite (single) + sensors | 0.97 | 1.00 | 0.94 | 1.00 | 0.47 |
| gemini/gemini-3.1-flash-lite (single) | 0.47 | 0.68 | 0.59 | 0.59 | 0.41 |
| ollama/moondream:1.8b (single) | – | 0.50 | 0.50 | 0.50 | 0.50 |

Table 5: Balanced accuracy by task, scaled set.

| Held-out task | Bal. acc. | False-succ. |
|---|---|---|
| stack | 0.50 [0.50, 0.50] | 100% [100, 100] |
| grasp | 1.00 [1.00, 1.00] | 0% [0, 0] |
| button | 0.66 [0.59, 0.74] | 68% [53, 83] |
| peg | 0.50 [0.50, 0.50] | 100% [100, 100] |
| push | 0.50 [0.50, 0.50] | 0% [0, 0] |

Table 6: Sensor classifier trained on four tasks, tested on the held-out one.

| Noise strength | Bal. acc. | False-succ. |
|---|---|---|
| 0 | 0.97 [0.96, 0.99] | 3% [1, 5] |
| 0.5 | 0.90 [0.87, 0.93] | 10% [6, 14] |
| 1 | 0.89 [0.85, 0.92] | 11% [7, 16] |
| 2 | 0.82 [0.78, 0.86] | 18% [13, 24] |
| 4 | 0.74 [0.70, 0.78] | 20% [14, 25] |
| 8 | 0.66 [0.61, 0.71] | 32% [25, 38] |
| 16 | 0.61 [0.56, 0.65] | 41% [34, 48] |
| 32 | 0.50 [0.45, 0.54] | 54% [47, 61] |

Table 7: Sensor classifier vs. noise strength (1 = default cheap hardware).

| Model | stack | grasp | button | peg | push |
|---|---|---|---|---|---|
| gemini/gemini-3.1-flash-lite | 11.3 | 0.0 | – | – | – |
| ollama/moondream:1.8b | 16.0 | 16.0 | 3.7 | 16.0 | 16.0 |

Table 8: Millimeters to fool (largest failure judged success; mean of 2 scenes).

| Judge | Also says success |
|---|---|
| gemini/gemini-3.1-flash-lite | 9/41 |
| ollama/moondream:1.8b | 15/15 |
| sensor classifier | 6/115 |

Table 9: Fooling scenes re-judged by other judges.

# 5 From judging to acting

The study above judges single snapshots. A robot agent decides many times per task and can act to find out. We built one: an LLM plans over five skills (move, grasp, lift, place, release) in the style of SayCan and Inner Monologue [5, 6]; MuJoCo WASM runs the Panda in a browser; damped least-squares IK executes. The same cheap sensors check each skill as it runs: a grasp must close both fingertip switches with effort above 5 N, a slip turns the switches off, and a placement lowers under a force guard until the wrist force rises. All experiments in this section use 80 seeded tasks (stack, tower, sort, zone), a scripted planner (no API calls, so runs are free and repeatable), up to three replans, and true simulator state for scoring. Failures are injected in the first attempt: a slip while carrying, the target moved just before placing, the cube knocked away right after release, or seeded random chaos.

**Who decides "done".** Table 10 compares decision rules. An OpenCV detector that fits each cube's 3D pose from color silhouettes stands in for the camera (the robot then runs open loop, without touch). Sensors catch failures during the task but are blind after release: alone they declared 80 of 80 knocked-off episodes a success. Camera and sensors together had 0 of 360 false successes.

| Failure | Camera | Sensors | Both | Simulator |
|---|---|---|---|---|
| None | 98 [91, 99] | 99 [93, 100] | 99 [93, 100] | 100 [95, 100] |
| Slip | 76 [66, 84] | 99 [93, 100] | 99 [93, 100] | 99 [93, 100] |
| Target moved | 50 [35, 65] | 98 [87, 100] | 98 [87, 100] | 100 [91, 100] |
| Knocked off | 75 [65, 83] | 0 [0, 5] | 79 [69, 86] | 79 [69, 86] |
| Chaos | 93 [85, 97] | 95 [88, 98] | 94 [86, 97] | 96 [90, 99] |

Table 10: True success (%, 95% Wilson interval) by decision rule and failure; 80 episodes per cell, 40 for target moved.

**Planning from what the camera sees.** By default the skills aim at true cube poses. In perceived-state mode they aim at the detector's estimates, refreshed before every attempt. Over 443 looks the median position error was 2.4 mm (mean 4.7 mm, worst cube 184 mm). Table 11 shows that no rule's true success changed significantly: grasps close on the cube and placements stop on contact, which absorbs millimeter errors.

| Decider, failure | n | Perceived | True state | McNemar |
|---|---|---|---|---|
| Simulator, none | 80 | 100 [95, 100] | 100 [95, 100] | 0/0, 1.000 |
| Sensors, none | 80 | 99 [93, 100] | 99 [93, 100] | 1/1, 1.000 |
| Both, none | 80 | 93 [85, 97] | 99 [93, 100] | 0/5, 0.063 |
| Both, chaos | 80 | 90 [81, 95] | 94 [86, 97] | 0/3, 0.250 |

Table 11: True success (%) with perceived and true state on the same tasks; McNemar counts only-perceived/only-true correct, and $p$.

**Touch the result before saying done.** Sensors go blind at release, so we added an active check in the spirit of interactive perception [7]: a *press probe* lowers the closed gripper onto each top cube and checks, from the noisy wrist force and the arm's own height, that the top face sits where the goal puts it. A router runs checks cheapest first: sensors; the camera only if the sensors pass; the probe only if the camera disagrees. Table 12 pools all five failure types. Our first probe checked the height where the robot had put each cube and missed towers rebuilt on a knocked-off middle cube; the current one uses the height the goal implies (both runs are in the repository). On knocked-off cubes false success fell from 80 of 80 with sensors alone to 2 of 80 with the probe; the remaining misses were towers of the right height in the wrong order, which a press cannot see and a camera can. The router matched sensors with the probe on true and false success and used 1.19 camera checks per episode against 2.07 for camera and sensors together, at 30.7 against 30.4 simulated seconds per episode.

**A 6-axis wrist sensor.** We replayed every committed attempt with a 6-axis force/torque sensor at the wrist (the replayed single-axis traces matched the originals exactly) and gave it the same noise model per axis. Task-agnostic features (contact onset, peak change, direction relative to the approach, steadiness after onset, torque) replace the single-axis wrist features; switches and effort stay. On the scaled set balanced accuracy rose from 0.89 to 0.96 (McNemar $p < 0.001$; Table 13), and held-out-task accuracy rose on stack and peg but not on button or push. On the 100-sample set the logistic-regression gain was not significant ($p = 0.481$). With more noise than the default, the advantage disappears: the intervals overlap from twice the default on.

| Strategy | True success | False success | False, knocked | Sim s | Camera | Probes |
|---|---|---|---|---|---|---|
| Sensors | 76 [71, 80] | 23 [19, 28] | 100 [95, 100] | 22.6 | 0.00 | 0.00 |
| Sensors + probe | 95 [92, 97] | 1 [0, 2] | 3 [1, 9] | 41.5 | 0.00 | 1.92 |
| Camera + sensors | 93 [90, 95] | 0 [0, 1] | 0 [0, 5] | 30.4 | 2.07 | 0.00 |
| Router | 95 [92, 97] | 1 [0, 2] | 3 [1, 9] | 30.7 | 1.19 | 0.45 |

Table 12: Checking strategies over all five failure types (360 episodes each): true and false success (%, Wilson interval), false success when the cube was knocked off after release, simulated seconds, camera checks and probes per episode.

| Classifier | Single axis | 6-axis F/T | McNemar |
|---|---|---|---|
| log. reg. | 0.89 [0.85, 0.92] | 0.96 [0.94, 0.98] | 41/12 |
| boosted trees | 0.89 [0.85, 0.92] | 0.96 [0.94, 0.98] | 36/6 |
| held out: stack | 0.50 | 0.89 | |
| held out: grasp | 1.00 | 1.00 | |
| held out: button | 0.66 | 0.54 | |
| held out: peg | 0.50 | 0.86 | |
| held out: push | 0.50 | 0.50 | |

Table 13: Single-axis vs. 6-axis wrist sensing, scaled set. McNemar: samples only 6-axis / only single-axis got right.

**Can a language model read a touch trace?** Each small-set trace, with the evaluation's noise and printed at 10 Hz, went to a text-only model (Table 14). One-shot adds one labelled success/failure pair of the same task from the training seeds. Zero-shot, the model is worse than the sensor classifier; given one example pair it is not significantly different from the classifier and clearly better than the camera VLM on the same samples. Reading a touch trace, given a reference, is within reach of a free text model; reading the gap from an image is not.

| Reader | n | Balanced acc. | False succ. | vs sensor clf. |
|---|---|---|---|---|
| gpt-oss-120b (0-shot) | 100 | 0.68 [0.58, 0.78] | 0.18 | 9/27, 0.005 |
| gpt-oss-120b (1-shot) | 96 | 0.91 [0.83, 0.97] | 0.10 | 6/2, 0.289 |

Table 14: A text LLM judging from the sensor trace alone (small set). Last column: McNemar against the sensor classifier.

**Harder instructions.** Twenty hand-written tasks test the planner on relational references, cubes that start stacked, impossible requests (where the plan schema allows a structured refusal) and distractor phrasing (Table 15). The LLM planner refused every impossible task and no possible one; the suite proved easier than intended and does not rank planners.

| Condition | Relational | Reorder | Impossible | Distractor | Correct |
|---|---|---|---|---|---|
| scripted | 5/5 | 4/5 | 5/5 | 5/5 | 95 [76, 99] |
| scripted-chaos | 5/5 | 4/5 | 5/5 | 5/5 | 95 [76, 99] |
| llm | 5/5 | 4/5 | 5/5 | 5/5 | 95 [76, 99] |
| llm-chaos | 4/5 | 4/5 | 5/5 | 5/5 | 90 [70, 97] |

Table 15: Harder suite: correct per family (goal met, or refusal for impossible tasks). Scripted rows plan from the hand-written goal.

# 6 Limitations

Simulation only; nothing here is validated on hardware, and simulated touch and force are cleaner than real ones even after degradation. The failures are synthetic and drawn from five tasks we designed, so the sensor features fit those tasks; leave-one-task-out results show they do not generalize. The minimal pairs are hard by construction, so VLM accuracy here says little about natural failure distributions. Free-tier quotas limited coverage for several models, and local models ran on downscaled images. VLM confidence is self-reported text, not a probability. We used one prompt per task and did not tune prompts per model. Two scenes per task in the attacker is small. Fusion is cross-validated on the test set rather than trained on separate VLM-labelled data. In the agent experiments the planner is scripted, failures are injected by teleporting cubes or opening the gripper, the probe thresholds were set once and not tuned, and perceived-state mode still gives the planner a scene description from the simulator.

# 7 Reproducibility

Code (MIT), the dataset with images, sensor traces, simulator states, ground truth and every model answer (CC BY 4.0), and a one-command script are public. Cached answers make reruns free.

# References

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